



Lego Microscope

Lego Ideas has a new product which is a fully functional microscope made of Lego
<http://dgit.in/LegoScope>

Air-conditioner cum cooler

The IcyBreeze cooler can be used too cool your drinks and also as an air-conditioner
<http://dgit.in/IcyBreeze>



Colonisation would require planetary biodomes to sustain life and implement research on new planets..

is Kepler-186f about 493 light years away. Not an easy trip to make into an unknown part of space.

As citizens of the solar system we have spent the bulk of the last 50 years making remarkable advances in localised navigation. Simply put we have a better chance at making it to Mars than to Kepler-186f. And looking further beyond nearby planets, we know that the moons of Jupiter such as Europa offer a far more realistic opportunity for humanity to explore and colonise. These seemingly farfetched flights of fancy are no longer just within the realm of science fiction but widely recognised among the scientific community as an essential part of humanity's survival.

With the release of The Millennium Project's "State of the Future" report indicating the massive challenges facing humanity, NASA's chief scientist Dennis Bushnell even committed on record that "the entire ecosystem is crashing" and given the massive growth rates in population humanity will "need at least three planets" if it is going to survive. The *State of the Future* report does indicate that the threats brought on by population, climate change, limited

resources and over-consumption are a real issue and not simply the premise of an epic Christopher Nolan movie. The answer in such a scenario can reasonably be to advance our ambitions within our local system first and let the farther reaches of space proceed as a Plan B.

Terraforming or Colonizing?

Amongst the vast community of scientists the terms terraforming and colonisation can evoke radically polarised reactions. While



That could be Titan in a few hundred years

colonisation is seen to be an essentially necessity driven act by humanity as a means to further our existence in space, terraforming becomes a greatly ethical and moral dilemma. The problem with terraforming planets is akin to the Prime Directive as envisioned by Star Trek. After all, terraforming is a radical interference with the natural state of a world. So the question arises - should humanity alter Nature? Some believe why not? Since we already have.

The Earth of today versus that just one hundred years ago is a radically different planet. From atmospheric composition to the nature of the biosphere, if we were to travel back in time, the planet would be almost alien for most people around the world. The lack of super-structures, electronic and mechanical technologies and the relative sparseness of resources would make Earth a challenge for most modern humans. The ethical dilemma then isn't "whether we should" but "how should we" - keeping in mind all the mistakes we've made due to lack of foresight and greed with our own planet.

Terraforming a planet is clearly beyond the scope of current technology but not